

Question 4 of 5

$$a) P(\text{getting bubble}) = \frac{1}{1000}$$

$$n = 8000 \quad p = \frac{1}{1000}$$

large n , small p

Using Binomial to Poisson approximation

$$\lambda = np = \frac{8000}{1000} = 8$$

$P(\text{getting fewer than 7 items with bubbles})$

$$= P(X < 7)$$

$$= e^{-\lambda} \sum_{x=0}^6 \frac{\lambda^x}{x!}$$

$$= e^{-8} \left(1 + 8 + \frac{8^2}{2} + \frac{8^3}{3!} + \frac{8^4}{4!} + \frac{8^5}{5!} + \frac{8^6}{6!} \right) \quad (5)$$

$$= \cancel{e^{-8}} \underline{\underline{0.313}} \quad (1)$$

b)

$$X \sim N(\mu, \sigma^2)$$

$$P(X \leq 8) = 0.9332$$

$$\Rightarrow \Phi\left(\frac{8-\mu}{\sigma}\right) = 0.5 + 0.4332$$

$$\Rightarrow \frac{8-\mu}{\sigma} = 1.5$$

$$\Rightarrow 8 - \mu = 1.5\sigma$$

$$\Rightarrow \boxed{\mu = 8 - 1.5\sigma} \quad (1)$$

$$P(X \geq -10) = 0.9938$$

$$\Rightarrow P\left(\frac{X-\mu}{\sigma} \geq \frac{-10-\mu}{\sigma}\right) = 0.9938 = 0.5 + 0.4938$$

$$\Rightarrow \Phi\left(-\frac{(-10-\mu)}{\sigma}\right) = 0.5$$

$$\Rightarrow 1 - \Phi\left(\frac{-10-\mu}{\sigma}\right) = 0.5 + 0.4938$$

$$\Rightarrow \Phi\left(\frac{-10-\mu}{\sigma}\right) = 0.5 + 0.4938$$

$$\Rightarrow \frac{10+\mu}{\sigma} = 2.5$$

$$\Rightarrow \boxed{\mu = -10 + 2.5\sigma} \quad (2)$$

for (1) and (2)

$$8 - 1.5\sigma = -10 + 2.5\sigma$$

$$18 = 4\sigma \Rightarrow$$

$$\boxed{\sigma = \frac{18}{4} = 4.5} \quad (4)$$

$$\Rightarrow \therefore \boxed{\mu = 8 - 1.5 \times 4.5} \quad (3)$$

$$= 1.25$$

$$c) \quad f(x) = \frac{\alpha^p}{\Gamma(p)} e^{-\alpha x} x^{p-1} \quad x > 0$$

$$M(t) = E(e^{tx})$$

$$= \int_0^{\infty} \frac{\alpha^p}{\Gamma(p)} e^{-\alpha x + tx} x^{p-1} dx$$

$$= \frac{\alpha^p}{\Gamma(p)} \int_0^{\infty} e^{-x(\alpha-t)} x^{p-1} dx$$

~~$\alpha = t$~~

$$\text{put } x(\alpha-t) = z$$

$$dx = \frac{dz}{\alpha-t}$$

$$M(t) = \frac{\alpha^p}{\Gamma(p)} \int_0^{\infty} e^{-z} \left(\frac{z}{\alpha-t} \right)^{p-1} \frac{dz}{\alpha-t}$$

$$= \frac{\alpha^p}{(\alpha-t)^p \Gamma(p)} \int_0^{\infty} e^{-z} z^{p-1} dz$$

$$= \frac{\alpha^p}{(\alpha-t)^p \cancel{\Gamma(p)}}$$

$$= \left(\frac{\alpha}{\alpha-t} \right)^p$$

$$= \left(1 - \frac{t}{\alpha} \right)^{-p}$$

(2)

$$E(x) = \left(\frac{dM(t)}{dt} \right) \bigg|_{t=0} = -p \left(1 - \frac{t}{\alpha} \right)^{-p-1} \left(-\frac{1}{\alpha} \right) \bigg|_{t=0}$$

$$= \frac{p}{\alpha}$$

(2)

$$E(X^2) = \frac{d^2 M(t)}{dt^2} \Big|_{t=0}$$

$$= \frac{p}{\lambda} (p-1) \left(1 - \frac{t}{\lambda}\right)^{p-2} \left(-\frac{1}{\lambda}\right) \Big|_{t=0}$$

$$= \frac{p(p-1)}{\lambda^2}$$

$$\text{Var}(X) = E(X^2) - E(X)^2$$

$$= \frac{p(p-1)}{\lambda^2} - \frac{p^2}{\lambda^2}$$

$$= \frac{p^2}{\lambda^2} + \frac{p}{\lambda^2} - \frac{p^2}{\lambda^2}$$

$$= \frac{p}{\lambda^2}$$

(3)